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# 5. WP3 – Generalized KS Transformations and N-Body Regularizations

**Expanding KS Beyond the Two-Body Template:** Numerous generalizations of the Kustaanheimo–Stiefel (KS) regularization have been developed to tackle more complex gravitational systems and multiple bodies. These extensions introduce *higher-dimensional "oscillator-like" phase spaces* with extra coordinates (often quaternions or spinors) and constraints, aiming to regularize collision singularities and reveal hidden symmetries. Here we survey this "KS-like zoo" – from modified single-Kepler transforms to multi-body regularizations – and assess whether their defining constraints can be understood as momentum map conditions for some group action (*Q1–Q3*).

- **Generalized KS for One Kepler Problem:** The classical KS map (4D spinor space → 3D Kepler space) has been reformulated and extended in various ways. For example, Breiter & Langner (2017) developed a *canonical KS transformation with an arbitrary preferred direction*, instead of the usual assumption that one coordinate axis (e.g. $x$-axis) is special [1] [2]. This yields a coordinate-free quaternionic KS mapping valid for any chosen "defining vector" in $\mathbb{R}^3$ [3] [4]. Such generalizations preserve the essential property that KS connects the 4D isotropic harmonic oscillator to the 3D Kepler problem [5]. Indeed, *every KS-type map "lifts" the Kepler motion to a simple oscillator on a higher-dimensional bundle* – a fact highlighted by the *Hopf fibration* geometry underlying KS [6] [7]. Recent studies recast the KS mapping in modern symplectic terms: Ferrer and Crespo (2018) showed that the KS transform can be viewed as a symplectic reduction by an $S^1$ action, where the famous KS **bilinear constraint is exactly the zero level of a momentum map** [8]. In other words, the extra phase (spinor angle) in $T^*\mathbb{R}^4$ *is a cyclic coordinate whose conjugate momentum (a certain quadratic combination of $u,p_u$) remains constant; fixing that momentum and quotienting out the $S^1$ fiber recovers the physical 3D motion* [8]. *(Interestingly, Ferrer & Crespo find* two *independent $S^1$ symmetries – "twin" bilinear integrals – in the KS-regularized oscillator, corresponding to two momentum maps* [9] [10]*. Only one is needed for generic Kepler orbits; both $S^1$ actions become relevant in degenerate cases like rectilinear orbits* [11] [12]*.) These results confirm the conjectural picture for the* 2-body Kepler prototype: *one of the KS algebraic constraints (specifically, the bilinear relation linking the 4D and 3D coordinates) is indeed a momentum map condition* [8]*. By contrast, the familiar norm relation $|u|^2 = r$ (which ties the 4D spinor length to the physical radius) behaves as a second-class constraint and is not directly a moment-map level in the standard formulation. Researchers have also extended the KS idea to* Kepler-like problems with additional fields*: e.g. the MIC-Kepler system (Kepler with a central Dirac monopole) can be incorporated by a generalized momentum mapping into the $so(2,4)$ symmetry algebra* [13]*. In fact, performing a Hopf-type reduction on a 4D oscillator naturally yields a 3D Coulomb system* with a magnetic monopole field [14] *– the monopole's vector potential emerges as the curvature of the $S^1$ bundle (the Berry connection of the KS fibration). This is a concrete example of how a nontrivial* bundle curvature *(here, the Dirac monopole) appears as a "magnetic" force in the reduced space (*Q4*), consistent with the general conjecture that upstairs integrability may become downstairs non-integrability via a curvature term.

- **Regularizing Multiple Collision Systems (Two-Center and Beyond):** Beyond a single Kepler orbit, KS-type techniques have tackled systems with *multiple gravitational centers or multiple bodies*. A

classic example is the **two-center (Euler) problem** – one particle moving under two fixed inverse-square forces. In the planar case, G.D. Birkhoff (1915) constructed a clever coordinate transform to *simultaneously regularize both collision singularities* at the two centers [15]. His mapping, later understood as a composition of complex Möbius transformations and a Levi-Civita squaring, sends a double-collision trajectory to infinity in a regular way [16] [15]. This idea was extended to the *spatial two-center problem* by Stiefel & Waldvogel (1970): they announced a 3D analog of Birkhoff's map using quaternions [17]. Waldvogel (1974) elucidated that the spatial construction is analogous to the planar one, but now leveraging the **Hopf fibration** (a map $S^3 \to S^2$) in place of the complex squaring [17]. Essentially, a *pair of KS transforms* (one per center) can be composed to regularize two collisions at once, with appropriate constraints to ensure the physical coordinates match up. In modern terms, this suggests an **$S^1 \times S^1$ action** might underlie the double-center regularization: one circle for each Levi-Civita/Hopf fiber. However, in practice the composition is achieved through explicit coordinate formulas [15], and an explicit momentum-map interpretation of the two-center constraints has not been spelled out in the literature (the transforms are defined analytically rather than as a symplectic quotient by a two-torus, as far as sources indicate). Still, the success of the two-center KS mapping reinforces the geometric theme: higher-dimensional rotations or phase shifts can "untangle" multiple singularities simultaneously by lifting to a suitable bundle (here, an $S^1$-invariant 3-sphere embedded in $\mathbb{H}=\mathbb{R}^4$ [18] [19]).

- **N-Body Regularization (Aarseth–Zare and Heggie):** The ultimate generalizations of KS aim to handle *any pairwise collisions in an N-body system*. Pioneering work by Aarseth & Zare (1974) provided an **eight-dimensional regularization of the general three-body problem**, based on applying a KS transform to one selected binary pair [20]. By augmenting the phase space and enforcing additional constraints, their method achieved **regularization of both binary collision singularities** ($R_{13}\to0$ or $R_{23}\to0$) in the three-body system [21]. Importantly, they found the extended formulation also improved the behavior of solutions during close triple approaches (near-simultaneous collisions) [22]. The trick was to introduce *redundant variables* representing the multiple relative position vectors and then impose relations among them. For instance, in a three-body triangle with bodies 1,2,3, one can treat the three relative vectors $(\mathbf{r}_{12}, \mathbf{r}_{31})$ *as independent coordinates in $\mathbb{R}^9$ and then impose the* ***closure constraint*** *$\mathbf{r}_{12}+\mathbf{r}_{23}+\mathbf{r}_{31}=0$* to tie them together. Aarseth & Zare essentially did this (following ideas of Broucke & Lass 1973) [23], then applied a KS (quaternion) mapping to one of the pairwise vectors. The result is a **phase space of dimension 8** (an increase from the original 6 for the reduced 3-body problem) with several algebraic constraints (norm constraints from KS, plus the closure condition). Heggie (1974) generalized this to a **global N-body regularization** that, in principle, can regularize *collisions between any pair* of bodies [24]. His approach was to chain multiple KS mappings along a spanning tree of the $N$ bodies: first introduce $N-1$ independent relative-coordinate vectors (connecting the bodies in a chain or tree), then replace each 3-vector by a 4-vector (quaternion) via KS. This yields a **$T^*!\mathbb{R}^{4(N-1)}$ extended space** with $(N-1)$ separate KS constraints (each like $|u_i|^2 = \text{(relative distance)}$ for a link in the chain) and additional linear constraints ensuring the *redundant coordinates are consistent with a single physical configuration*. In the three-body case, for example, one ends up with 3 KS-type constraints and 3 closure equations (though not all are independent) as noted in our conjectural setup. Heggie's formalism showed that the **equations of motion can indeed be written in a form regular at all collision events** [24]. However, the cost is a proliferation of constraints and the need to switch "charts" or re-label the tree when different pairs become the closest (to maintain numerical stability [25]). From a geometric standpoint, each KS quaternion introduces an **$S^1$ gauge symmetry** (the

arbitrary phase of the 4D spinor). If all $(N-1)$ quaternions were truly independent, one might expect an underlying $T^{N-1}$ torus action with a momentum map given by the collection of bilinear forms ${|u_i|^2 - R_i}$ (each set to zero). In practice, though, the interdependence of the quaternions (via closure constraints and re-labeling conditions) complicates any simple group interpretation. *We found no evidence in the literature that a single unified momentum map encompassing all multi-body KS constraints has been formulated.* The KS conditions in the N-body context are typically enforced by augmenting the Hamiltonian with Lagrange multipliers or using Dirac's theory of constrained systems, rather than by invoking a clear Noether symmetry. Nevertheless, the **analogy to gauge theory is evident**: the extra phase coordinates in multi-KS regularization are non-physical (gauge) degrees of freedom introduced to smooth out singular forces, and they could potentially be fixed by a choice of section (just as one fixes a gauge) or quotiented out. Identifying a concrete $G$-action whose momentum map yields *all* these constraints remains an open problem in the reviewed literature – a gap exactly in line with our conjecture's target.

- **"Bargmann–Symmetrization" Perspective:** A complementary viewpoint on these regularizations is provided by the *Bargmann extension* (sometimes called the **collision-blowup via time dilation**). In Bargmann's approach, one embeds the dynamical system into a higher-dimensional space that includes the time as an additional coordinate, so that the motion in the extended space is geodesic (free or harmonic) [26]. For the Kepler problem, this is famously achieved by adding one extra coordinate (besides time) to make an *O(4,2)*-symmetric 5-dimensional space in which the trajectories of the 3D Kepler motion lift to straight lines [27] [28]. The KS transformation is deeply connected with this: it provides the spatial part of that lift (the Hopf fibration mapping the physical trajectory to a circle in $S^3$), while a rescaled time (the "anomaly" parameter) accounts for the motion along the fiber. In the presence of multiple bodies or perturbations, similar extended-state strategies exist in theory (e.g. lifting the entire $N$-body system to a geodesic flow on a null cone in a $(3N+2)$-dimensional Bargmann manifold), but they require many integrals of motion to carry out explicitly. For instance, the Aarseth–Zare 3-body regularization can be seen as a **Hamiltonian time-transform plus coordinate lift**: it uses the constant center-of-mass momentum and energy (six first integrals) to reduce the problem, then introduces a new time variable $s$ (with $dt/ds$ proportional to an appropriate function of inter-body distances) to regularize the remaining dynamics [29]. The end result is effectively a *8D phase space with a modified symplectic form* ("extended by $dt\wedge dE$") in which the flow is analytic through collisions. While powerful for analysis and numerics, this approach is more implicit and does not always yield the clear geometric insight of a principal bundle or momentum map structure. Still, it underlines a common theme: **adding extra dimensions can trade a problematic "downstairs" force (like $1/r^2$ singularities or Coriolis terms) for a benign "upstairs" geometry**. The extra dimensions typically carry a curvature or fictitious force that re-enters the reduced equations as a "magnetic term" (e.g. the monopole field in the Hopf reduction [14]). In the restricted three-body problem, for example, one can extend the phase space by an angle coordinate to absorb the induced Coriolis force, yielding an equivalent description with a background magnetic field [30]. We will explore these curvature effects in WP4, but it is worth noting here that many KS-type regularizations implicitly introduce such connections. Whenever a rotation or phase is added (be it an $S^1$ Hopf fiber or an $SU(2)$ frame rotation), any *nontrivial holonomy* in that fiber bundle manifests as a dynamic gauge field after reduction, potentially explaining why integrals exist "upstairs" but break "downstairs."

**Summary of Generalized KS Constructions:** We compile in Table 5.3 a summary of notable KS-based regularizations beyond the basic 2-body case, highlighting the extended phase space, constraint structure, and any known group/momentum-map interpretation:

| System / Transform | Extended Phase Space (dim $M$) & New Vars. | Group Symmetry $G$ (if identified) | Constraints Introduced | Momentum Map? |
|---|---|---|---|---|
| **Kepler (2-body)** – KS (1965) [4] | $T^*!\mathbb{R}^4$ (spinor $u\in \mathbb{H}$) with $s$-time | $S^1$ (Hopf fiber rotation) [8] | $ | u |
| **Kepler + Monopole** – MIC-Kepler [14] | $T^*!\mathbb{R}^4$ + gauge coord. (for $U(1)$ field) | $S^1$ or $SO(3)$ (hidden $so(2,4)$ symmetry) | $ | u |
| **Two-Center (planar)** – Birkhoff (1915) [15] | $T^*(\mathbb{C}^2)$ via two sequential LC maps | (Implicit $S^1\times S^1$ actions in steps) | Two independent Levi-Civita constraints (one per center) | *Not explicit:* achieved via analytic mapping (Moebius + $z^2$) [16] |
| **Two-Center (spatial)** – Stiefel–Waldvogel (1971) [17] | $T^*(\mathbb{H} \times \mathbb{H})$ (two quaternions) | $S^1\times S^1$ (two Hopf fibers) | Two KS constraints ($ | u_1 |
| **Restricted 3-Body (spatial)** – e.g. KS regularization [31] | $T^*(\mathbb{R}^4)$ extended by one quaternion (regularize planet–sun) | $S^1$ (like 2-body KS) + $ \mathbb{Z} _2$ (time reversal) | $ | u |
| **General 3-Body** – Aarseth–Zare (1974) [20] | $T^*(\mathbb{R}^4 \times \mathbb{R} ^3)$ (one quaternion $u$ + one 3-vector) | $S^1$ (phase of $u$) possibly + (rotations)* | KS constraint $ | u |
| **Global $N$-Body** – Heggie (1974) [24] | $T^*(\mathbb{R} ^{4(N-1)})$ (chain of $N-1$ quaternions $u_i$) | $\approx T^{N-1}$ (phases $ \phi_i$ for each $u_i$) | $ | u_i |

| System / Transform | Extended Phase Space (\$M\$) & New Vars. | Group Symmetry \$G\$ (if identified) | Constraints Introduced | Momentum Map? |
|---|---|---|---|---|
| **"Multifold Kepler" integrable family** – Cordani *et al.* (1990) [32] | \$T^*(\mathbb{R}^4)\$ or higher (depending on system) | \$SO(4)\$, \$SU(2)\$,... (extended Runge–Lenz sym.) | Various (e.g. one monopole-like and one KS-like constraint for closed orbits) | **Yes:** e.g. 2D and 3D harmonic oscillators yield momentum maps generating closed orbits [33] (Bertrand's theorem cases) |

<small>*Table 5.3:* Selected generalizations of the KS regularization, showing their extended phase space, added symmetries/constraints, and whether the constraints can be interpreted as moment map conditions. "Momentum map" here means an identified conserved quantity whose zero level enforces the KS constraint, associated to a continuous symmetry.</small>

**Findings:** Many of the above constructions *partially* fulfill the idea that "KS constraints = momentum map zero-levels on a principal bundle." In particular, the standard KS map for the two-body problem can be cleanly understood in this way [8], and even admits multiple commuting \$S^1\$ symmetries in the regularized 4D oscillator [9]. Generalizations that involve *independent* KS mappings (e.g. separate fibers for separate pairs) tend to carry a torus (\$T^k\$) of gauge phases, but when physical constraints link the fibers, the symmetry is reduced or no longer Abelian. We did not find an existing publication that unifies all pairwise KS constraints of the full 3-body or \$N\$-body problem into a single compact momentum-map formalism – this appears to be an open gap. The ingredients are all present (multiple \$S^1\$ actions, closure constraints reminiscent of a \$\mathbb{R}^3\$ momentum map for translations, etc.), but a synthesis is missing. This gap is exactly where our conjecture in *WP1/WP5* points: it suggests that a judicious choice of bundle (perhaps a nontrivial \$S^1\$ or \$SU(2)\$ bundle over shape space) might capture **all** necessary constraints as moment maps, with a curvature form encoding the complexity of the unreduced dynamics. The literature so far provides strong **hints** – e.g. the appearance of Dirac monopole fields after Hopf reduction [14], and the improved integrability of multi-body systems "upstairs" [25] – but the full conjecture for a *"generalized KS = momentum map, 3BP"* is, to our knowledge, not yet realized in published form. This WP3 survey thus maps out what is known and where the conjecture stands: the 2-body case is confirmed, various extended KS schemes exist for more bodies, and the role of bundle curvature as a "hidden force" is recognized, but a complete group-theoretic regularization of the three-body problem remains an open challenge.

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